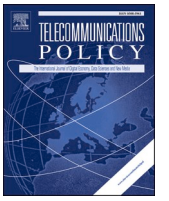
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Planned obsolescence and smartphone replacement: Empirical evidence on the Italian market

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ABSTRACT

In an era of rapid technological change, planned obsolescence has significantly shaped consumer behaviour, especially in the smartphone industry. This study examines how material, functional, psychological, and economic obsolescence affect smartphone lifespans, using original survey data from Italian consumers. While material remains relevant, its influence diminishes in newer smartphone generations, where software-related limitations and repairability gain prominence. Functional obsolescence is positively associated with device longevity, as many users tolerate performance issues, particularly those with high technological engagement. By contrast, psychological and economic obsolescence remain strong drivers of early replacement. Consumer attitudes interact with different types of obsolescence: users with higher technological engagement are more tolerant of material and functional decline, while pro-repair consumers are less likely to replace devices due to material wear. Furthermore, there is a shift in replacement behaviour over time: for older smartphones, being second-hand significantly shortens lifespan, whereas for newer devices, user behaviour, brand perceptions, and repair attitudes play a more influential role—highlighting a growing consumer awareness around sustainability and functionality. These findings provide insights for both policymakers and manufacturers, highlighting the need for sustainable business practices and robust regulatory frameworks to counteract planned obsolescence and promote longer-lasting products in an environmentally conscious market.

1. Introduction

In today's rapidly evolving technological landscape, where innovation cycles are fast and the push toward sustainable practices grows, the phenomenon of planned obsolescence—where manufacturers deliberately design products with shorter lifespans to drive repeat purchases—has come under significant scrutiny. This strategy, deeply rooted in consumer markets, has historically driven economic growth and spurred technological advancement by promoting research, development, and cost reductions (Hamblet, 2023). By fostering demand for newer models and upgrades, planned obsolescence has shaped consumer behaviour and propelled industries forward. However, the economic benefits are increasingly offset by rising social and environmental costs, as consumers face higher expenses from frequent product replacements and a diminished ability to repair or extend the lifespan of their goods. Indeed, many manufacturers prioritize sales over durability, reducing repair options and thereby frustrating consumers, who perceive this as a manipulative strategy to keep them within a cycle of continuous consumption. As a result, consumer durables have paradoxically

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become non-durables, embodying shorter lifespans and contributing to a faster turnover of goods (Hartl et al., 2023).

The implications of shorter product lifespans extend beyond individual consumption, influencing global resource demands and environmental sustainability. Planned obsolescence encourages the rapid consumption of durable goods, leading to accelerated resource extraction, increased energy use, and heightened waste generation. The linear nature of this model, whereby products quickly transition from use to disposal, intensifies the environmental footprint, adding to global challenges around resource scarcity, waste management, and carbon emissions (Niklewicz-Pijaczyńska et al., 2021).

The smartphone industry stands out as a notable example, often cited by consumer advocacy groups for its high turnover rates and extensive use of programmed obsolescence tactics. With over 4.5 billion users globally (Deloitte, 2022), smartphones have become essential tools for communication, productivity, and entertainment, yet they are replaced on average every two years (Makov & Fitzpatrick, 2021). Manufacturers have been accused of employing various strategies to incentivize frequent upgrades, including limiting software updates for older models, designing phones with non-removable batteries and other hard-to-repair components, and continuously introducing new models with minor updates (AGCM, 2018; Altroconsumo, 2020). Such practices impose a heavy toll not only on consumers' finances but also on the environment, as each device demands significant raw materials, energy for production, and eventually contributes to electronic waste. The ecological impact of the smartphone industry is particularly concerning, given the extraction of critical raw materials, the energy-intensive production processes, and the global accumulation of electronic waste (Cordella et al., 2021).

In response to these growing concerns, regulatory bodies have begun to address planned obsolescence through legal and policy measures. France has led the way with legislation banning planned obsolescence, followed by similar initiatives within the European Union. The EU has implemented a range of policies, including the Ecodesign directive, the EU Ecolabel system, and the Right to Repair directive, aimed at extending product lifespans and reducing environmental impacts through improved repairability and sustainability standards (Maggiolino, 2019; Malinauskaite & Erdem, 2021; Šajn, 2022). Alongside these regulatory developments, consumer attitudes are shifting as well; research indicates that users are slowing the rate at which they replace their smartphones and increasingly demand devices that last longer and offer more accessible repair and upgrade options (European Commission, 2018). This shift in consumer preference reflects a growing awareness of sustainability and a desire to counteract the disposability of digital devices (Barros & Dimla, 2021).

Against this backdrop, understanding the impact of planned obsolescence in the smartphone industry is crucial for both businesses and policymakers, as it can inform strategies to enhance sustainability while meeting consumer demands. This paper investigates how various types of obsolescence affect the lifetime of smartphones, with a particular focus on whether these dynamics have shifted with the latest generation of devices. Specifically, the analysis addresses the following research questions: how do different forms of planned obsolescence—material, functional, psychological, and economic—influence smartphone replacement decisions, and how are these effects shaped by consumers' attitudes and usage practices? Have these patterns changed over time with the diffusion of newer generations of smartphones?

The analysis focuses on Italy, which offers a compelling case for investigating planned obsolescence and consumer behaviour due to its exceptionally high smartphone penetration rate, widespread direct ownership of devices, and a market context that highlights consumer-driven replacement dynamics. As detailed in Section 3, Italy stands out among European countries for both the intensity of smartphone use and the structure of its mobile market.

Leveraging an original dataset on Italian consumers, this analysis explores the underlying factors that prompt smartphone replacement, revealing that all four types of obsolescence—material, functional, psychological, and economic—play distinct roles. While visible wear (material obsolescence) remains relevant, its influence appears to diminish in newer smartphone generations, where software-related limitations and repairability gain prominence. Interestingly, functional obsolescence is positively associated with device longevity, as many users tolerate performance issues, particularly those with high technological engagement. By contrast, psychological and economic obsolescence remain strong drivers of early replacement. The study also shows that consumer attitudes interact with different types of obsolescence: users with higher technological engagement are more tolerant of material and functional decline, while pro-repair consumers are less likely to replace devices due to material wear. Furthermore, we identify a shift in replacement behaviour over time: for older smartphones, being second-hand significantly shortens lifespan, whereas for newer devices, user behaviour, brand perceptions, and repair attitudes play a more influential role—highlighting a growing consumer awareness around sustainability and functionality.

Our study contributes in different ways to the literature on planned obsolescence, particularly in the context of smartphone lifespans and consumer replacement behaviour. First, while existing research has classified different forms of planned obsolescence—such as functional, psychological, material, and economic obsolescence (Guiltinan, 2009; Packard, 1961; Proske et al., 2016, pp. 1–8)—our study advances this discussion by empirically examining how these factors interact in shaping smartphone replacement cycles. We provide a nuanced understanding of how these mechanisms operate simultaneously rather than in isolation. Furthermore, previous research has often treated economic and psychological obsolescence as distinct drivers of product replacement (Makov & Fitzpatrick, 2021; Van Nes & Cramer, 2006). Our findings suggest that these factors increasingly overlap in the smartphone market, particularly as manufacturers use trade-in programs and repair cost structures to create the perception that upgrading is the economically rational choice. From an empirical perspective, our study utilizes original survey data on young Italian consumers, an underexplored demographic in planned obsolescence research. Unlike prior studies that primarily rely on aggregate market trends or experimental settings, our dataset provides individual-level insights into actual replacement decisions. In this context, we can quantify the role of different types of obsolescence as well as the interaction between consumer attitudes – towards technology, innovation, aesthetics and repairability - and obsolescence. This adds an important behavioural dimension to the study of planned obsolescence, moving beyond firm strategies to incorporate consumer agency in replacement decisions. Finally, we analyse smartphone replacement over time (two

generations of devices per respondent), in order to understand how consumers' behaviour has changed across different generations of products and what factors area associated with the replacement.

These findings provide valuable insights for policymakers and industry leaders. They underscore the need for enhanced regulatory focus on durability and repairability standards and offer businesses opportunities to innovate in ways that align with sustainable consumption patterns. The paper is organized as follows: Section 2 provides a review of the literature on planned obsolescence, Section 3 outlines the methodology and data collection process, Section 4 discusses the empirical strategy and presents the variables of the analysis, while Section 5 illustrates the results. Finally, section 6 concludes by exploring strategic and regulatory measures to mitigate the impacts of planned obsolescence and extend the lifespan of electronic devices.

2. Planned obsolescence and the smartphone industry

2.1. The concept of planned obsolescence

The concept of planned obsolescence traces back to the early 20th century, arising as a strategy for boosting sales during economic downturns (Slade, 2007). General Motors (GM) was one of the first companies that developed such a strategy. In 1913, GM executive Alfred Sloan introduced the electric starter, rendering hand-cranked cars obsolete. Sloan further spurred demand by instituting annual styling changes to GM vehicles, encouraging frequent trade-ins (Slade, 2007). Around the same time, the Phoebus Cartel, an alliance of light bulb manufacturers, artificially reduced bulb lifespans to increase sales (Valant, 2016). The term "planned obsolescence" itself first appeared in Bernard London's 1932 pamphlet, proposing the deliberate design of products with finite lifespans to stimulate consumption and mitigate the effects of the Great Depression (Hamblet, 2023). Later, industrial designer Brook Stevens popularized the concept in 1954, defining it as a tactic to foster demand for newer, supposedly superior products sooner than necessary (Valant, 2016). By the 1960s, however, planned obsolescence faced criticism, particularly in Vance Packard's *The Waste* Makers (1961), which condemned the practice as a wasteful strategy aimed at fostering consumer dissatisfaction (Slade, 2007). These historical cases illustrate the deliberate design and marketing strategies that laid the groundwork for more systematic attempts to define and classify different types of obsolescence.

Obsolescence is a multifaceted phenomenon, influenced by industry practices, consumer behaviour, and regulatory interventions. The early typologies, such as those by Packard (1961), were foundational in identifying functional, quality-based, and psychological obsolescence. However, these early classifications have been further developed to reflect contemporary technological and business dynamics. Guiltinan (2009) highlights the strategic nature of obsolescence, distinguishing between physical constraints (minimal repairability) and technological mechanisms (incremental feature upgrades to drive replacement cycles). Burns (2010) expands this by incorporating aesthetic and economic factors, recognizing that perceived product value is equally important as actual performance deterioration. However, these classifications primarily focus on manufacturer-driven obsolescence, neglecting consumer-driven factors such as evolving preferences, brand perception, and sustainability awareness. This gap is increasingly relevant in the smartphone industry, where brand loyalty and technological engagement influence replacement behaviours (Makov & Fitzpatrick, 2021). Furthermore, Manier and Mugge (2022) suggest that consumer dissatisfaction with the perceived quality of products, rather than actual performance decline, plays a key role in replacement behaviour, emphasizing the role of expectation management and emotional attachment in prolonging product lifespans.

Recent research suggests that planned obsolescence is increasingly embedded in technological platforms, where software updates, app compatibility, and service discontinuation contribute to perceived functional obsolescence (Makov & Fitzpatrick, 2021; Manier & Mugge, 2022). This further reinforces the shift from purely hardware-driven obsolescence to system-driven constraints that limit product longevity. Prakash et al. (2020) highlight that while software-driven obsolescence is often framed as an unavoidable technological necessity, it also serves as a strategic tool to push consumers towards device upgrades.

When discussing the issue of planned obsolescence, the distinction between planned obsolescence and natural obsolescence is crucial. Planned obsolescence implies the intent—or at least awareness—of the manufacturer to shorten the lifespan of products (Sapmaz Veral, 2022). It often takes the form of functional, psychological, economic, or aesthetic obsolescence, where a product is either physically constrained in its usability or perceived as outdated due to marketing and social influences (Guiltinan, 2009; Makov & Fitzpatrick, 2021). In contrast, natural obsolescence refers to the inevitable, unplanned deterioration of products over time due to wear and tear, material degradation, or changing environmental factors.

A significant theoretical contribution to the economic literature on planned obsolescence came from Bulow (1986), who provided a formal framework to understand the strategic incentives behind the practice. His work challenges the assumption that firms in a competitive market would naturally produce goods with an efficient level of durability, ensuring an economically optimal lifespan. Instead, Bulow demonstrates that firms with market power can benefit from deliberately reducing product durability, creating conditions where planned obsolescence is a profit-maximizing strategy. By shortening product lifespans, firms increase the frequency of repeat purchases or shift consumers toward rental or subscription models, ensuring a steady revenue stream.

The economic implications of planned obsolescence are further explored by Orbach (2004), who introduces the concept of "durapolists"—monopolists in the durable goods market. Unlike producers of perishable goods, durapolists face declining demand over time because consumers can reuse durable products, thereby limiting their incentives for frequent repurchases. This challenge is compounded by the existence of second-hand markets and the reluctance of consumers to pay premiums for extended durability. To counteract these market frictions, durapolists may deliberately reduce product lifespans, ensuring that replacement cycles remain active. Strategies such as limiting repairability, discontinuing spare parts, and implementing software-driven obsolescence further reinforce this cycle.

While planned obsolescence is often framed as an industry-driven phenomenon, recent studies emphasize the role of consumer behaviour. Manier and Mugge (2022) argue that perceived obsolescence—where consumers *believe* their product is outdated—drives premature replacement, even when devices remain fully functional. This aligns with Jaeger-Erben et al. (2023), who show that consumers increasingly demand durability but remain influenced by social and technological pressures. Additionally, Alzayidi (2024) argues that shifting sustainability attitudes and repairability expectations are reshaping consumer choices. As a result, planned obsolescence is no longer a one-sided strategy; firms must navigate changing consumer priorities while maintaining profitability. The interplay between firm-driven obsolescence and consumer-driven replacement behaviour suggests that long-term sustainability policies must balance innovation incentives with regulatory intervention. The growing demand for repairability, longevity, and sustainability reflects an evolving market context that could redefine product replacement cycles (Barros & Dimla, 2021; Makov & Fitzpatrick, 2021; Prakash et al., 2020).

2.2. *The evolution of planned obsolescence in the smartphone industry*

The smartphone industry has become a key focus of discussions on planned obsolescence, as it embodies both hardware and software-driven mechanisms that accelerate device replacement cycles (Makov & Fitzpatrick, 2021). While early concerns about obsolescence centered on appliances and automobiles, smartphones have introduced new dynamics due to their rapid innovation cycles, high consumer engagement, and strong dependencies on software updates and service compatibility (Manier & Mugge, 2022). Consumer organizations and researchers have long cited smartphones as one of the most susceptible product categories to obsolescence practices, as users often report performance declines, software slowdowns, and incompatibilities following system updates (Makov & Fitzpatrick, 2021; Valant, 2016). This aligns with studies showing that planned obsolescence in smartphones is increasingly psychological and software-driven rather than hardware-based (Manier & Mugge, 2022).

Unlike traditional durable goods, smartphones are subject to both manufacturer-driven and consumer-driven replacement incentives. Firms use incremental feature improvements, aggressive marketing, and operating system updates to encourage users to upgrade more frequently than strictly necessary (Jaeger-Erben et al., 2023). At the same time, consumers exhibit high brand loyalty, leading to habitual replacement cycles that reinforce market control by dominant firms (Scalability or Durability, Jaeger-Erben 2023).

Globally, smartphone replacement cycles have lengthened in recent years, shifting from an average of 2.4 years in 2013 to 3.7 years by 2022. There are, however, important differences across countries. For example, in the United States, the average expected life span (replacement cycle length) of consumer and enterprise smartphones was 2.67 and 2.54 years, respectively, in 2023,[1] while in the UK almost 30 % of surveyed consumers use their smartphone up to two years and 41 % up to 4 years.[2] Although these figures suggest a shift toward longer smartphone lifetimes, firms continue to deploy obsolescence strategies to counteract this trend. For instance, research shows that software updates often lead to slower device performance, reinforcing functional obsolescence perceptions (Prakash et al., 2020). In addition, battery degradation, limited repairability, and non-removable components contribute to hardware-driven obsolescence (Sabbaghi & Behdad, 2018).

Smartphone manufacturers increasingly rely on software-driven obsolescence, where updates limit compatibility with older devices, reduce system efficiency, or introduce new technical requirements (Makov & Fitzpatrick, 2021; Prakash et al., 2020). This strategy is often framed as technological progress, but consumer advocacy groups argue that it serves as a deliberate tactic to push upgrades. One of the most publicized cases of software-driven obsolescence occurred in 2017, when Apple was found to have intentionally slowed down older iPhones following iOS updates (Romm, 2020). The company later admitted to throttling performance to compensate for aging batteries, but critics argued this manipulated consumer behaviour by reinforcing the perception that older devices were “too slow” to function properly (Makov & Fitzpatrick, 2021). Similarly, in 2018, Samsung was fined by the Italian Competition Authority (AGCM) for misleading consumers into installing updates that caused significant performance degradation (AGCM, 2018). These cases highlight the fine line between legitimate software improvements and planned obsolescence tactics.

While planned obsolescence is often examined through the lens of software updates and consumer perceptions, hardware constraints play a fundamental role in shaping product longevity. The physical durability of smartphones is influenced by several design choices, including fragile materials, non-replaceable batteries, and the growing use of non-modular components that hinder repairability (Sabbaghi & Behdad, 2018). Many manufacturers integrate batteries that are difficult or costly to replace, solder critical components directly onto the motherboard, and employ proprietary screws or adhesives that complicate repairs (Makov & Fitzpatrick, 2021). These practices not only accelerate device turnover but also pose a direct challenge to circular economy initiatives (Prakash et al., 2020). The shift toward thinner and lighter designs—often driven by aesthetic considerations and competitive marketing strategies—further exacerbates these issues, making devices more fragile and less adaptable to future upgrades (Jaeger-Erben et al., 2023). Moreover, limited access to spare parts and manufacturer-imposed repair restrictions reduce repair feasibility, reinforcing short replacement cycles and discouraging third-party repair services (Valant, 2016). The durability of smartphones is not solely determined by technological advancements; rather, it is shaped by a strategic balance between enhanced performance and controlled obsolescence to maintain sales momentum (Manier & Mugge, 2022). Research highlights that planned hardware constraints, combined with systemic barriers to repairability, are among the key drivers of premature smartphone replacements. Addressing these issues requires not only increased consumer awareness and regulatory intervention but also fundamental changes in product design strategies that

[1] https://www-statista-com.eu1.proxy.openathens.net/statistics/619788/average-smartphone-life/.

[2] https://single-market-economy.ec.europa.eu/document/download/f5cffe4b-317f-429b-a597-f154fb33d69b_en?filename=SWD_2023_101_1_EN_impact_assessment_part3_v3.pdf.

prioritize repairability, modularity, and material sustainability (Alzayidi, 2024; Barros & Dimla, 2021; Hahn et al., 2020). Without these shifts, the industry's focus on frequent upgrades will continue to undermine both consumer trust and environmental sustainability, perpetuating unnecessary e-waste and resource depletion (Makov & Fitzpatrick, 2021).

Due to their small size (Slade, 2007), the environmental footprint of smartphones is particularly high. According to Deloitte (2021), on average, a new smartphone produces about 85 kg of carbon emissions within its first year of operation, with 95 % of these emissions resulting from manufacturing processes such as raw material extraction and transportation. The same study estimated that only in 2022, smartphones would generate a total of 146 million tons of CO_2. Furthermore, smartphones are playing an increasingly significant role in the overall greenhouse gas emissions (GHG) generated by the ICT sector. This surge is happening at such a rapid pace that it has the potential to surpass the GHG emissions produced by desktops, laptops, and displays (Cordella et al., 2021). As most of the carbon footprint of smartphones comes from the manufacturing process (Deloitte, 2021), the most effective way to decrease the environmental impact of these devices would be to increase their lifespan, hence slowing down the resource loop (Bocken et al., 2016) by reducing planned obsolescence. Beyond emissions, resource depletion and e-waste accumulation are critical concerns. The production of smartphones relies on rare earth metals and critical raw materials, often extracted under environmentally harmful and socially problematic conditions (Prakash et al., 2020). The increasing e-waste problem further exacerbates these concerns, as smartphones contribute to the fastest-growing category of waste worldwide (Hahn et al., 2020). Research indicates that extending smartphone lifespans by just one year could significantly reduce global e-waste and carbon emissions (Sabbaghi & Behdad, 2018). This suggests that policies aimed at improving repairability, recyclability, and durability are essential to mitigating the environmental consequences of smartphone production and disposal.

In line with these considerations, consumers have recently started to exhibit a shift in preferences towards durability and a certain intolerance for perceived obsolescence with regards to their smartphones. For instance, the *Behavioural study on consumers' engagement in the circular economy* conducted by the European Commission (2018) shows an increasing desire for longer-lasting durable goods, including smartphones, and a significant willingness to pay for better reparability. Corroborating these findings is a study carried out by McKinsey & Company (2021), which presents after-sale services as one of the main reasons behind users' dissatisfaction in the telecommunication industry. Furthermore, evidence shows that consumers worldwide have reduced the replacement rate of their smartphones, increasing their average life cycle (see Fig. 1). These data suggest that consumer preferences are changing, and new opportunities arise for companies who want to find new profitable ways to meet the needs of their customers.

The increasing resale and second-hand market for smartphones also suggests a broader shift in consumer attitudes toward sustainability and longevity (Alzayidi, 2024).

Governments have started to address planned obsolescence through regulatory interventions and policy measures. France was the first country to introduce legislation explicitly prohibiting planned obsolescence through the law 2015–992, which carries significant penalties, including a two-year prison sentence and fines of up to €300,000 or 5 % of a company's turnover. Other European countries, such as Italy and Spain, have pursued similar efforts, focusing on consumer protection, warranty extensions, and transparency in software updates. In 2017, the European Parliament adopted a resolution on programmed obsolescence, calling for an independent testing system, legal protection for whistleblowers, and greater transparency in product longevity (Maggiolino, 2019).

Several EU-level initiatives have also been introduced to combat planned obsolescence.

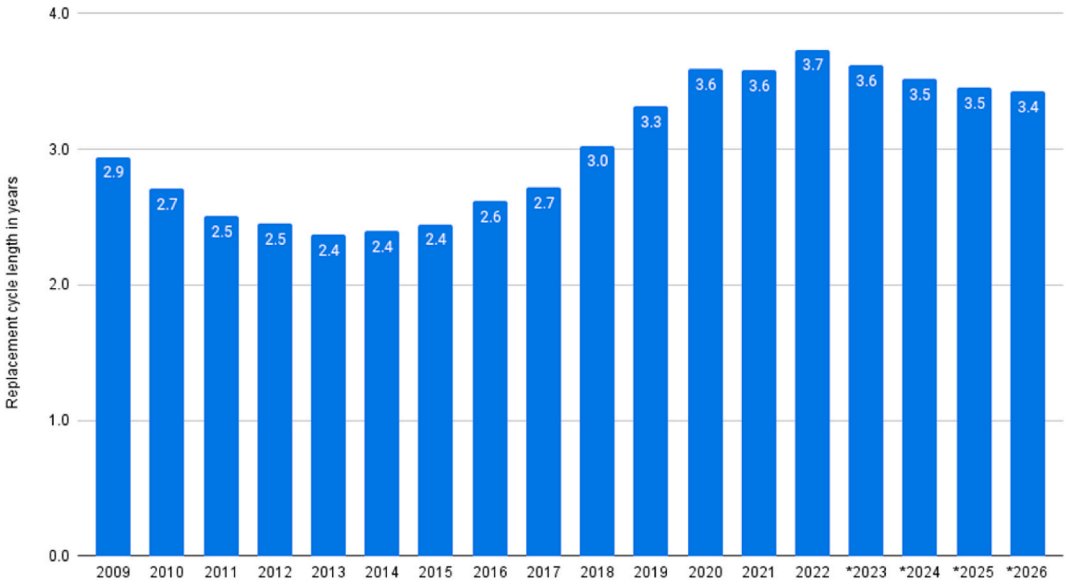


Fig. 1. Worldwide Smartphone Life Cycle
*Estimated values.
Source: https://www.sellcell.com/blog/how-often-do-people-upgrade-their-phone-2023-statistics/

- The Ecodesign Directive, which establishes minimum durability and repairability requirements for electronic products.
- The EU Ecolabel, a voluntary labelling scheme that promotes waste and CO_2 reduction during manufacturing.
- The Right to Repair Directive, aimed at reducing electronic waste by requiring manufacturers to provide spare parts and repair options for an extended period (Malinauskaite & Erdem, 2021; Šajn, 2022).

At the global level, initiatives such as extended producer responsibility (EPR) programs and stricter recycling mandates are gaining traction (Hahn et al., 2020). In some cases, companies have pre-emptively adjusted their business models, offering trade-in programs, refurbished products, and subscription-based models to align with changing consumer expectations and regulatory trends (Alzayidi, 2024).

Despite growing regulatory scrutiny, proving planned obsolescence remains a challenge, especially in the smartphone industry. Establishing conscious intent by manufacturers to shorten product lifespans is complex, as many forms of obsolescence emerge at the intersection of technological progress, consumer behaviour, and market forces. Some scholars argue that functional obsolescence is often a byproduct of genuine innovation rather than an explicit attempt to render older devices unusable (Makov & Fitzpatrick, 2021). Others suggest that psychological obsolescence, which stems from marketing, peer pressure, and shifting design aesthetics, plays an equally significant role in driving replacement decisions (Manier & Mugge, 2022). Moreover, recent research challenges the notion that obsolescence is purely manufacturer-driven. Jaeger-Erben et al. (2023) emphasize that consumer expectations have shifted, with many users demanding newer models and perceiving "old" devices as outdated, even when they remain fully functional. Similarly, Prakash et al. (2020) highlight that while consumers often blame manufacturers for obsolescence, their own purchasing behaviour reinforces short product cycles.

Ultimately, planned obsolescence must be understood within a broader technological, economic, and regulatory context. While shorter product lifespans may benefit manufacturers in the short term, rising consumer awareness, sustainability concerns, and regulatory changes are shaping new market dynamics. Companies must navigate a delicate balance between innovation, durability, and economic incentives. As a matter of fact, companies serve as problem solvers, as their fundamental purpose is to provide innovative solutions to unaddressed needs. On the other hand, sustainability has emerged as a key driver of innovation, alongside digitalization and evolving consumer preferences (Jørgensen & Pedersen, 2018). Firms that embrace circular economy models, modular designs, and extended repair services can turn sustainability into a competitive advantage, responding to both regulatory pressures and shifting market expectations. In this context, the transition to a more sustainable future is not just a challenge but a strategic opportunity. This study contributes to the ongoing debate on planned obsolescence by integrating both manufacturer strategies and consumer-driven resistance to obsolescence. By focusing on the smartphone industry, it highlights how consumer attitudes, regulatory developments, and repairability initiatives are reshaping the replacement decision process. Additionally, the study offers empirical insights into how different forms of obsolescence interact with consumer engagement, shedding light on the balance between perceived and actual product longevity.

2.3. Planned obsolescence practices applied to smartphones

To examine planned obsolescence within the smartphone industry, this study adopts Proske et al.'s (2016) classification, identifying key types of obsolescence.

— *Material/qualitative obsolescence:* Accelerated product deterioration due to insufficient material quality or component capacity, leading to premature aging.
— *Functional obsolescence:* Obsolescence driven by evolving technical or functional requirements, such as the interoperability of hardware and software.
— *Psychological obsolescence:* Consumers' subjective perception of aging due to changing fashion, social trends, or consumption patterns.
— *Economic obsolescence:* Loss of product functionality when maintenance or repair costs exceed the price of replacement.

Material obsolescence occurs when a product's physical degradation is accelerated by suboptimal material quality or component limitations, resulting in premature aging. In the smartphone context, material obsolescence manifests through common technical failures in components like batteries, glass displays, back covers, LCD screens, and overall performance degradation (Clemm et al., 2020; Cordella et al., 2021; Prakash et al., 2020; Watson et al., 2017). For example, frequent issues with batteries losing charge capacity and screens cracking due to design choices can prompt users to replace devices. While difficult to attribute definitively to manufacturers, practices like using fragile glass backs and maximizing screen-to-body ratios, which increase vulnerability to damage, indicate an implicit drive towards faster replacements (Clemm et al., 2020; Cordella et al., 2021; Hahn et al., 2020). Additionally, failures in components such as non-damaged but malfunctioning LCD screens can lead to replacement when repair options are limited, thus potentially linking material failures with economic obsolescence.

Functional obsolescence arises as technical requirements evolve, making existing products less effective. Smartphones, for instance, may lose usability as manufacturers discontinue software support or design updates that slow older models. Some examples include.

- Prematurely ending software updates for older models, often attributed to cost considerations.
- Software updates tailored for the latest models, which may degrade the performance of older versions.
- Increased storage demands associated with new software updates, limiting the ability to install updates on older devices.

- Absence of expandable storage options, preventing users from increasing storage capacity as needed.

Additionally, application developers may stop supporting older software versions, furthering functional obsolescence, even if not directly instigated by the smartphone manufacturer (Proske et al., 2016, pp. 1–8). These factors contribute to early replacement, as users seek devices compatible with new software and applications.

Psychological obsolescence stems from consumers' changing perceptions and preferences, often driven by shifts in fashion, technology trends, and social influences. Unlike functional obsolescence, psychological obsolescence is largely subjective, involving a consumer-driven desire for newness. Manufacturers, however, actively encourage this mindset through marketing strategies, product cycles, and limited feature enhancements that maintain consumer interest in the latest models. Strategies include minor annual updates that capitalize on consumers' preferences for staying current, incremental feature introductions to stimulate demand for new releases, and occasionally withholding technological advances to facilitate subsequent product launches. Service contracts that subsidize frequent upgrades also fuel psychological obsolescence, encouraging consumers to replace devices sooner. While telecom providers often offer these upgrade incentives, manufacturers also contribute, such as through Apple's iPhone Upgrade Program available in the U.S. (Proske et al., 2016, pp. 1–8).

Economic obsolescence occurs when the perceived cost of repair exceeds that of replacement. This type of obsolescence has both subjective and objective aspects, as consumers weigh repair costs against the anticipated remaining lifespan of their devices (Proske et al., 2016, pp. 1–8). Smartphone manufacturers contribute to economic obsolescence by designing products with non-removable batteries, extensive use of adhesives, and components that are difficult to access, making repairs costly. Examples include non-removable batteries that require professional repair services, adhesive use that complicates disassembly, and restricted access to official replacement parts. These design choices can limit repair options, effectively shortening product lifespans.

3. Methodology and data collection

Our empirical analysis explores how planned obsolescence affects consumers' attitudes toward smartphone replacement. In particular, this study addresses the following research questions.

- How do different forms of planned obsolescence—material, functional, psychological, and economic—influence smartphone replacement decisions, and how are these effects shaped by consumers' attitudes and usage practices?
- Have these patterns changed over time with the diffusion of newer generations of smartphones?

We investigate the issue with reference to the Italian market, which presents a compelling setting for several reasons. First, Italy has historically been one of the countries with the highest mobile phone penetration rates worldwide. According to World Bank data, by the end of 2022, Italy had 133 mobile subscriptions per 100 people, surpassing the average of high-income countries (131.3), the European Union (123.7), Germany and Sweden (125), and the United States (110.2). Second, Italy has an exceptionally high smartphone penetration rate of 97 %, placing its population among the heaviest smartphone users globally. The number of smartphone users has been steadily increasing, and the user base is projected to reach 53.92 million by 2029, setting a new record. Third, a recent Statista (2024) report highlights that 72 % of Italian consumers purchase their smartphones at full retail price, while 4 % buy second-hand, 9 % acquire them through mobile subscriptions, 5 % receive them as gifts from family or friends, and 2 % obtain them from employers. This high rate of direct ownership suggests that studying consumer-driven replacement decisions provides an accurate

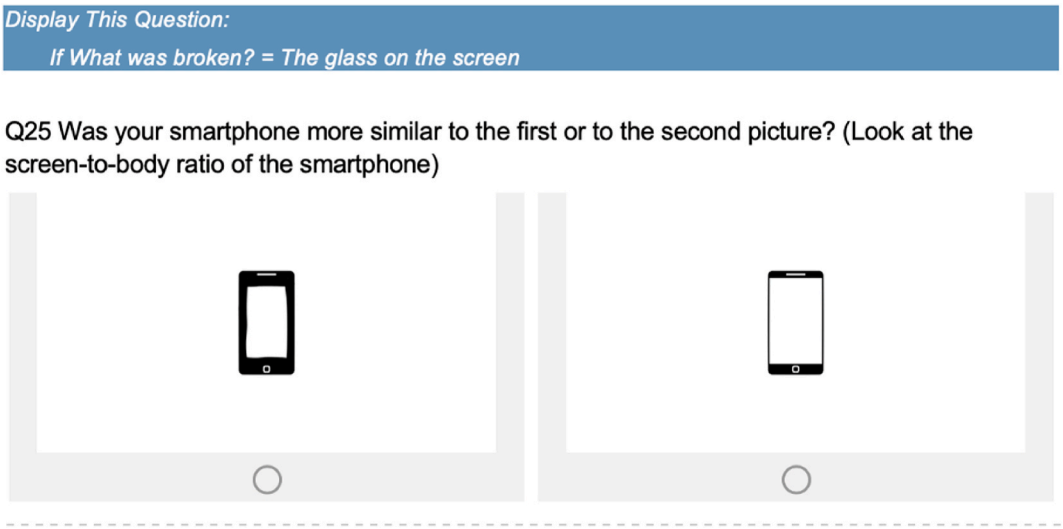


Fig. 2. Survey question no. 25 (screen-to-body ratio).

perspective on market dynamics.

To collect data on the motivations behind smartphone replacement, a customized survey was distributed to Italian smartphone users via social media and personal networks. Participants were invited to complete the survey in exchange for entry into a raffle for a €25 Netflix gift card, which helped to incentivize participation.

The survey consists of 48 questions organized into three sections to capture diverse information. The first section includes 16 questions focusing on respondents' demographics (e.g., age, gender, nationality), attitudes toward technology, product innovation, aesthetics, repair skills, and smartphone-related activities. Six of these questions specifically assess respondents' attitudes toward factors such as design, functionality, and repairability—key elements relevant to understanding psychological and economic obsolescence.

The second section centres on the smartphone model owned immediately before the respondent's current device, referred to as Smartphone A. This section contains 16 questions designed to investigate replacement motives and capture data on various obsolescence types. Most of them were constructed with the purpose of measuring the different types of planned obsolescence. An example of question that tries to measure material obsolescence is provided below (Fig. 2). As mentioned before, a high smartphone screen-to-body ratio increases the likelihood of damage to the glass on the screen and can thus be used as a measure of material obsolescence (see Clemm et al., 2020; Cordella et al., 2021; Hahn et al., 2020).

The third section mirrors the second but focuses on the smartphone model owned two generations prior to the respondent's current device, hereafter referred to as Smartphone B. This section's symmetry to the prior section enables a comparative analysis of obsolescence factors across different product generations, providing insight into potential shifts in consumer expectations and manufacturing practices. It is important to notice that we adopt a cross-sectional design with retrospective questions on two past smartphones (A and B).

In total, the survey gathered 381 responses. The sample includes participants aged 16–30 (44 %), with a gender distribution of 64 % female and 36 % male. In terms of occupation, employees make up 47 % of respondents, while students represent 28 %, ensuring a varied demographic representation in terms of age, occupation, and usage habits (see Table 2). The skewness of the sample toward individuals aged 16–30 can be justified by the demographic characteristics of smartphone users and their relevance to the topic under investigation. Younger individuals tend to be the most intensive and engaged users of smartphones, with high levels of technological proficiency and a greater reliance on mobile devices in their daily lives. According to various market reports (e.g., Deloitte, 2021), the 16–30 age group represents a particularly active segment in terms of smartphone usage, app engagement, and frequency of replacement. This demographic is also more exposed to and influenced by evolving trends in technology, brand marketing, and digital platforms, which are central to the study of planned obsolescence. Furthermore, younger users are often early adopters of new technologies and are more likely to experience and report on issues related to software updates, hardware degradation, and repairability—factors that are key to understanding the dynamics of obsolescence. As such, their inclusion provides valuable insights into the behavioural and attitudinal drivers of smartphone replacement. While this age concentration may limit generalizability across the entire population, it allows for a focused analysis of the consumer segment most directly affected by, and responsive to, the dynamics of planned obsolescence.

Preliminary analysis of the data reveals a notable difference between perceived and desired smartphone lifespans. Respondents generally expressed a desire for longer device longevity than they believe current smartphones deliver. Furthermore, Smartphone A shows a longer average lifespan than Smartphone B, suggesting evolving consumer preferences or changes in manufacturing that align with other industry findings (see Table 2). This trend supports the evidence showing a gradual extension in smartphone replacement cycles in recent years.

Given that some respondents did not own a smartphone two generations prior and considering that respondents' answers concerning products from a long time in the past may not be accurate enough, our main analysis studies the variables associated with the replacement of the most recent smartphone (smartphone A). However, given the increasing lifespan of smartphones over time highlighted in Table 2, we also look at the variables associated with the replacement of the less recent smartphone (smartphone B), mainly to identify possible changes in consumers attitude and behaviours towards smartphone replacement across different generations.

4. Empirical strategy

We investigate the variables associated with the lifetime of smartphone by estimating the following OLS model:

$$Lifetime = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 K_i + \beta_4 W_i + \varepsilon_i$$

Where X_i represents socio-demographic controls (e.g., age, gender, income, occupation), Z_i includes consumer attitudes and behaviours (e.g., technological engagement, repair attitudes, innovation perception), K_i accounts for different types of obsolescence (functional, material, psychological, economic), W_i controls for market factors (e.g., brand fixed effects) and ε_i is the error term. Given the cross-sectional nature of our data and the relatively small sample size, the objective of our empirical strategy is to identify significant patterns in consumer replacement behaviour, rather than to develop highly predictive models or establish causal relationships.

4.1. Independent variables

The focus of our study is primarily to measure the effect of planned obsolescence on the lifespan of smartphones. We therefore

include in the regressions four dummy variables accounting for the different types of obsolescence, which take value 1 if the specific obsolescence is present and 0 otherwise.

First, we control for m*aterial obsolescence*, which takes value 1 when the replacement stems from a damaged back cover or screen glass, but only when the back cover is made of glass, or the screen-to-body ratio is high respectively. As explained in section 3.1, these attributes notably align with characteristics usually associated with planned material obsolescence. We expect this variable to have negative correlation with the lifetime of smartphones.

Second, we include a variable accounting for f*unctional obsolescence*, which takes value 1 when respondents indicate that the device replacement was prompted by difficulties with new updates or insufficient storage. The expected influence of this variable on the product's lifespan remains uncertain, given that these reasons can manifest in both the initial and later stages of the smartphone's life cycle. In this context, deliberate planned obsolescence should be deemed applicable only when such difficulties occur within the initial two or three years of the product's lifespan. However, incorporating this condition into the construction of the variable would introduce a level of dependence on the dependent variable. Thus, this condition was omitted, and the dummy variable includes all instances of functional obsolescence in a broader sense. As such, we do not have any a-priori expectation on the effect of functional obsolescence on the lifetime of smartphones.

Third, we include a variable measuring *psychological obsolescence*, which takes value 1 when respondents express a desire or need for the latest model (both sentiments are regarded as equivalent in terms of susceptibility to psychological obsolescence). This variable also includes instances where consumers sign a contract with the manufacturer that allows upgrades to their smartphone after a specific period. Psychological obsolescence can also occur when consumers refuse to repair due to the belief that their smartphone's remaining lifespan is limited, as well as when they perceive that the device has undergone « too many repairs». This statement is based on the assumption that repairs can be carried out until the product is no longer fixable, making this case aligned with the perception of a brief remaining lifespan. Lastly, an individual indicating a desired lifespan lower than 4 years can also be deemed subject to psychological obsolescence, as we can assume that this belief influenced the replacement decision. We expect this factor to have a negative influence with the lifetime of smartphones.

Finally, *economic obsolescence* takes value 1 when respondents state they refrained from repairing their device due to the perception of high costs, also considering the remaining lifespan of the product. This category also includes instances where consumers claim that the device was not reparable, a reflection of manufacturers increasingly adopting new assembly techniques that hinder the disassembly and repair of smartphones, as elaborated in section 3.1, referencing Cordella et al. (2021). Table 1 summarises the obsolescence variables and the expected effects.

Besides looking at the role of obsolescence, we have included a set of control variables that refer to demographic characteristics, behavioural traits of the individuals concerning the attitude towards different smartphones' features, individual technological skills and smartphones' characteristics.

We first control for age and gender and for the attitudes of consumers towards technology, innovation, aesthetics and repair. In order to measure the attitude of consumers towards technology, a question in the questionnaire asked respondents to evaluate on a five-point scale their agreement with four statements: "I think technology is a fundamental part of our everyday life", "Technology is more than just a tool in support of our activities", "Technology is not a threat to society, "I like to have to integrate technology into my everyday life". These items reflect dimensions related to perceived usefulness, societal impact, and everyday integration of technology, and were inspired by constructs from the technology acceptance literature (e.g., Venkatesh et al., 2003). We performed a principal component analysis and extracted one component, which we label *tech_attitude*. Second, we measure the attitude of consumers towards product innovation in smartphones through a question, which asked respondents to evaluate on a five-point scale their agreement with five statements: "If I heard that a new smartphone version was available in the store, I would be interested enough to buy it", "In general, I am the first in my circle of friends to know about the latest new version of a smartphone", "I change my smartphone every year", "I like to have the best features available on my smartphone", "I like to have the best camera available to take good pictures". These items reflect interest in technological novelty, social positioning, and preference for high-performance features. They were

Table 1
Obsolescence variables.

Variable	Description	Expected effect
Materialobs	Replaced because the back cover was broken, and the back cover was made of glass	–
	Replaced because the glass on the screen was broken, if the cover was made of glass and the smartphone had a high screen-to-body ratio	
Functionalobs	Replaced because the smartphone was struggling with the new updates	?
	Replaced because the smartphone did not have enough storage	
Psychologicalobs	Replaced because the respondent wanted/needed the latest model	–
	Replaced because of a service contract with the manufacturer, which allows to upgrade the smartphone after a certain amount of time	
	Not repaired because the respondent perceives that it had been repaired too many times	
	Not repaired because the respondent perceives that the residual lifetime of the smartphone was already short	
	Desired lifetime lower than 4 years	
Economicobs	Not repaired because the respondent perceives it was too expensive to repair it	–
	Not repaired because the respondent perceives it was too expensive to repair it compared to its residual lifetime	
	Not repaired because it could not be repaired	

Table 2
– Variable description.

Variable Name	Description
age	Age of the consumer
gender	Gender of the consumer (male = 0, female = 1)
web_activity	The overall intensity of smartphone usage for web-related activities
filestorage_activity	The overall intensity of smartphone usage for doc/music-related activities
dailyusage	Number of hours per day the consumer uses the smartphone
new	Indicates whether the smartphone was purchased new (1) or second-hand (0)
tech_attitude	Consumer's engagement and attitude towards technology
innov_attitude	Consumer's engagement and attitude towards innovation
repair_attitude	Consumer's attitude towards repairing the smartphone
aesth_attidude	Consumer's sensitivity to the aesthetic/design aspects of the smartphone
techskills	Consumer's level of technological skills
materialobs	Perception of physical wear or obsolescence of the smartphone
functionalobs	Perception of functional degradation of the smartphone (e.g., performance issues)
psychologicalobs	Perception of psychological obsolescence (e.g., feeling the phone is outdated)
economicobs	Perception of economic inefficiency (e.g., repair costs, resale value)
Brand	Dummy variables accounting for Apple, Samsung, Other
Income	Dummy variables accounting for the level of income
Occ	Dummy variables accounting for the type of occupation

Table 3
– Summary statistics.

Variable	Obs	Mean	Std. dev.	Min	Max
lifetimea	381	3.318898	1.285797	0.3	8
lifetimeb	339	2.951622	1.161165	0.1	8
age	381	40.81365	17.09184	16	80
gender	381	0.6404199	0.4805083	0	1
web_activity	381	1.23e-08	1.771862	−9.328186	2.355871
filestorage_activity	381	−1.56e-10	1.045321	−3.43633	5.270937
dailyusage	381	4.818635	2.967818	0.1	17.6
new	381	0.8687664	0.3380996	0	1
tech_attitude	381	−2.54e-09	1.414514	−5.795358	2.638855
innov_attitude	381	4.48e-09	1.6907	−2.446641	6.310177
repair_attitude	381	−8.62e-09	1.15603	−4.555573	1.968225
aesth_attidude	381	3.74e-09	1.663289	−3.328772	4.744272
techskills	381	5.77e-09	1.603169	−4.7833	2.990735
materialobs	381	0.0472441	0.2124395	0	1
functionalobs	381	0.1574803	0.364732	0	1
psychologicalobs	381	0.2650919	0.4419625	0	1
economicobs	381	0.4041995	0.4913817	0	1
	381				
Categorical Variables	*Frequency*	*Percentage*			
Income					
Low	37	9.71			
Lower Middle	117	30.71			
Upper Middle	86	22.57			
High	28	7.35			
Very High	17	4.46			
Not Independent	96	25.2			
Occupation					
Employee	178	46.72			
Other	21	5.51			
Retired	23	6.04			
Self-employed	47	12.34			
Student	108	28.35			
Brand (Smartphone A)					
Apple	183	48.03			
Samsung	102	26.77			
Other	96	25.20			
Brand (Smartphone B)					
Apple	158	41.47			
Samsung	90	23.62			
Other	133	34.91			

developed to reflect innovation-seeking behaviour and are conceptually related to findings on technology early adopters (Venkatesh et al., 2003; Rogers, 2003; Roerich, 2004). We performed a principal component analysis and extracted one component, which we label *innov_attitude*. Third, in order to capture the importance of product design and aesthetic for consumers, a question asked respondents to evaluate on a five-point scale their agreement with five statements: "I care about the design of my smartphone", "Among my group of friends, I like to have the best smartphone", "I like my smartphone to be in line with my style", "I do not like to have an old-looking smartphone", "I do not like to have cracks on my screen or display". These items were inspired by prior research on product attachment and aesthetic degradation as a driver of product replacement (Cooper, 2004; Mugge, Schifferstein, & Schoormans, 2005). We performed a principal component analysis and extracted one component, which we label *aesth_attitude*. Finally, given that we are interested in understanding the replacement patterns, we consider the attitude of consumers towards the possibility of repairing smartphones instead of buying new ones. A question asked respondents to evaluate on a five-point scale their agreement with three statements: "If my smartphone is damaged, I will try to repair it", "If the cost of repair is too high from the original manufacturer (e.g., brand), I will ask to independent repair cafés", "If the cost of repair is too high compared to that of a new smartphone, I will buy a new smartphone". These items reflect willingness to repair and sensitivity to cost-related trade-offs, and are related to empirical findings on the decline of repair practices in consumer behaviour (McCollough, 2009). We performed a principal component analysis and extracted one component, which we label *repair_attitude*. We also include a set of interaction terms, as it is reasonable to argue that consumer attitudes moderate the effect of different types of obsolescence on smartphone replacement decisions. Specifically, we first interact *tech_attitude* with material and functional obsolescence, since consumers with a higher technological attitude may be more sensitive to material obsolescence or functional obsolescence. If these users prioritize performance and cutting-edge features, they may be less tolerant of wear and slowdowns, accelerating replacement behaviour. Then we interact *aesth_attitude* with psychological obsolescence: consumers who emphasize aesthetics and design may be more likely to replace their smartphones frequently due to psychological obsolescence. This occurs when the perceived modernity of a device, rather than its actual performance, determines replacement timing. Finally, we introduce the interaction between *repair_attitude* and material obsolescence: consumers with a strong repair mindset may exhibit a different response to material obsolescence compared to those with lower repair attitudes. Rather than replacing a worn-out phone, they may attempt to fix or refurbish it, leading to longer lifespans.

An important variable when studying replacement behaviours is the level of usage of smartphones (Huh & Kim, 2008; Shih & Venkatesh, 2004). We control for this, leveraging upon four questions in the questionnaire. First, a question asked respondents to evaluate on a five-point scale a set of statements related to the level of their technical skills: "I consider myself able to use the latest smartphone versions", "I like to use my smartphone for more activities other than traditional calls, texts and social media", "I always find new ways to use my smartphone", "If a new feature is available, I have to learn how to use it". We performed a principal component analysis and extracted one component, which we label *techskills*. Second, we control for the type of activities individuals perform with their smartphones through a question investigating the intensity on a five-point scale. The activities are taking pictures, writing emails, web browsing, managing documents, listening to music, using social media, calling/texting. We performed a principal component analysis and extracted two components: the first – *web_activity* - is mostly explained by browsing the web and using social media, while the second – *filestorage_activity* - is mostly explained by managing documents and listening to music. Third, we control for the daily usage in terms of hours (*dailyusage*) and for the fact that the smartphone was originally new or second-hand through the dummy *new*, which takes value 1 if the smartphone was new and 0 otherwise.

Finally, we control for the brand of the smartphone and for the income and occupation level of the consumers. Since 75 % of the sample owns a smartphone by Apple or Samsung (see Table 4), we create a dummy variable for these two brands and one of the others. As far as income and occupation variables are concerned, we include a set of dummies to capture the potential effects of financial status and employment type on smartphone replacement behaviour. Income is categorized into multiple categories, distinguishing between lower, middle, and higher-income groups, while also accounting for individuals who are financially dependent (e.g., students supported by parents). Similarly, occupation dummies differentiate between employed, self-employed, unemployed, students, and retirees, allowing us to examine whether professional status influences the timing and reasons for device replacement.

A complete summary of the variables employed in the empirical model that includes the principal components is presented in Table 2, while Table 3 presents the summary statistics.

To assess the convergent and discriminant validity of the attitudinal constructs, we performed a principal component analysis (PCA) with varimax rotation on all relevant survey items (see Table A1 in the Appendix). Four components emerged, reflecting the theoretical dimensions of technology orientation, innovation orientation, aesthetic sensitivity, and repair attitude. Most items loaded strongly on their intended factor, with limited cross-loadings. For instance, items related to aesthetic preferences (e.g., aestheticstyle, aestheticdesign, aestheticpeers) and innovation (innovationnewversion, innovationknowledge, innovationfeatures) displayed high loadings on distinct factors, supporting the internal coherence of each construct. The reliability of the overall scale was high (Cronbach's alpha = 0.8465).

In addition, pairwise correlations among the resulting PCA-based factor scores remained moderate or low, further supporting the discriminant validity of the constructs (See Table 4). Pairwise correlations among the PCA-based attitudinal factors were generally moderate, supporting their distinctiveness. The highest correlation was observed between innovation and aesthetic attitudes ($r = 0.73$), suggesting a degree of conceptual overlap, yet retaining sufficient differentiation to justify separate constructs. As expected, age was negatively correlated with digital activity, daily usage, and technological skills, reflecting generational differences in smartphone engagement. Importantly, the obsolescence measures exhibited low correlations with attitudinal and behavioural variables, indicating they capture distinct dimensions of smartphone replacement motivations.

Finally, to assess multicollinearity among the independent variables, we computed the Variance Inflation Factors (VIF) for the full model, both in the case of smartphone A and in the case of smartphone B. All values were below the commonly accepted threshold of 5

Table 4
– Correlation matrix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
age (1)	1000														
gender (2)	0,000	1000													
web_activity(3)	−0,375	0,134	1000												
filestorage_activity (4)	−0,359	−0,097	0,000	1000											
dailyusage (5)	−0,196	−0,089	0,340	0,208	1000										
new (6)	0,117	0,065	0,064	−0,006	0,098	1000									
tech_attitude (7)	−0,001	−0,127	0,216	0,146	0,215	0,063	1000								
innov_attitude (8)	0,016	−0,087	0,229	0,152	0,242	0,116	0,357	1000							
repair_attitude (9)	−0,093	0,069	0,264	−0,078	0,092	−0,076	0,242	0,026	1000						
aesth_attitude (10)	−0,128	−0,106	0,259	0,182	0,260	0,081	0,313	0,730	0,107	1000					
techskills (11)	−0,167	−0,220	0,353	0,221	0,322	0,039	0,481	0,520	0,192	0,498	1000				
materialobs (12)	0,066	−0,117	−0,140	0,047	−0,128	0,013	−0,128	−0,099	−0,086	−0,118	−0,086	1000			
functionalobs (13)	0,118	0,084	0,000	−0,070	−0,037	−0,003	0,030	0,040	0,010	0,041	−0,030	−0,096	1000		
psychologicalobs (14)	−0,049	−0,046	0,094	0,095	0,203	0,040	0,102	0,166	0,040	0,135	0,112	0,034	−0,113	1000	
economicobs (15)	−0,168	−0,029	0,041	0,124	0,049	−0,028	−0,027	−0,134	0,026	−0,125	0,030	0,220	−0,356	−0,107	1000

(mean VIF = 2.09 for the model of smartphone A and mean VIF = 2.18 for smartphone B), indicating that multicollinearity does not pose a significant issue for our estimates.

5. Results

Table 5 displays the results of the analysis. In particular, Model 1 represents the basic model, Model 2 includes the dummy variables accounting for brands (Apple is the reference case), Model 3 controls for income and Model 4 is the full model including also dummy variables for occupational status.

Obsolescence factors play a significant role in determining smartphone lifespans. Starting with material obsolescence, we find a negative and significant coefficient, indicating that wear and tear significantly shorten the phone's lifespan, consistent with prior research. Material degradation, particularly visible or tactile wear, often encourages consumers to replace devices even when core functions remain unaffected (Cooper, 2004; Van Nes & Cramer, 2006). This effect is particularly pronounced for frequently used, highly visible products like smartphones. It is interesting to notice that material obsolescence loses significance when brand controls

Table 5
– Variables associated with smartphone's lifetime (recent smartphones).

VARIABLES	Model (1)	Model (2)	Model (3)	Model (4)
age	−0.00136	−0.00136	0.00285	0.00565
	(0.00452)	(0.00444)	(0.00565)	(0.00679)
gender	−0.216	−0.280**	−0.336**	−0.335**
	(0.133)	(0.129)	(0.141)	(0.148)
web_activity	−0.0805	−0.104**	−0.0975*	−0.0873*
	(0.0510)	(0.0506)	(0.0514)	(0.0520)
filestorage_activity	−0.0858	−0.107*	−0.0963	−0.0973
	(0.0632)	(0.0598)	(0.0627)	(0.0647)
dailyusage	0.0385*	0.0401*	0.0400*	0.0367
	(0.0230)	(0.0229)	(0.0228)	(0.0229)
new	0.234	0.498***	0.538***	0.550***
	(0.175)	(0.185)	(0.192)	(0.197)
tech_attitude	0.0695	0.0895*	0.0910*	0.0927*
	(0.0528)	(0.0492)	(0.0507)	(0.0505)
innov_attitude	−0.0369	−0.0592	−0.0637	−0.0744
	(0.0577)	(0.0576)	(0.0579)	(0.0579)
repair_attitude	−0.0732	−0.0572	−0.0647	−0.0584
	(0.0638)	(0.0622)	(0.0603)	(0.0623)
aesth_attitude	−0.0641	−0.0884	−0.0930	−0.108*
	(0.0616)	(0.0595)	(0.0607)	(0.0622)
techskills	−0.00331	−0.0174	−0.0241	−0.0134
	(0.0517)	(0.0503)	(0.0507)	(0.0509)
materialobs	−0.461**	−0.223	−0.128	−0.159
	(0.178)	(0.163)	(0.176)	(0.194)
functionalobs	0.563***	0.616***	0.576***	0.561***
	(0.195)	(0.190)	(0.194)	(0.194)
psychologicalobs	−0.677***	−0.575***	−0.585***	−0.597***
	(0.142)	(0.136)	(0.137)	(0.143)
economicobs	−0.424***	−0.420***	−0.446***	−0.403***
	(0.141)	(0.136)	(0.136)	(0.141)
tech_material_interaction	−0.0853	−0.198*	−0.193*	−0.209*
	(0.0971)	(0.102)	(0.101)	(0.113)
tech_functional_interaction	−0.244	−0.274	−0.263	−0.279*
	(0.182)	(0.170)	(0.168)	(0.163)
repair_material_interaction	0.0835	0.425**	0.513***	0.528**
	(0.209)	(0.194)	(0.195)	(0.208)
aesth_psychological_interaction	0.0488	−0.00191	0.0269	0.0538
	(0.0781)	(0.0736)	(0.0747)	(0.0776)
Samsung		−0.520***	−0.555***	−0.592***
		(0.167)	(0.174)	(0.183)
Other		−0.788***	−0.835***	−0.834***
		(0.168)	(0.173)	(0.177)
INCOME DUMMY VARIABLES	YES	YES	YES	YES
OCCUPATION DUMMY VARIABLES	YES	YES	YES	YES
Constant	3.405***	3.507***	3.332***	3.082***
	(0.291)	(0.284)	(0.352)	(0.447)
Observations	381	381	381	381
R-squared	0.184	0.238	0.248	0.257

Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.
Note: All models include dummy variables for income group, occupation, and smartphone brand. For reasons of space and because they are not central to the research questions, only the brand coefficients are reported. The reference category here is Apple.

are introduced. This suggests that brand choice may be absorbing part of the effect that material wear has on smartphone replacement decisions. One possible explanation is that different brands have varying levels of durability and material quality, which consumers take into account when purchasing their devices. Premium brands are often associated with higher material quality and longevity, reducing the independent effect of material wear on replacement behaviour. This aligns with findings from Jaeger-Erben et al. (2023), who emphasize the scalability vs. durability trade-off in consumer technology. Moreover, brand-specific consumer behaviour plays a role; for instance, Apple users may be more likely to replace their phones regularly due to software updates, trade-in incentives, or status signalling, regardless of actual material degradation. Additionally, if higher-end brands offer frequent trade-in programs or software support, users may upgrade before material obsolescence becomes a decisive factor. Finally, differences in repairability and second-hand market value across brands may also contribute to this result. Brands with better repairability options may see consumers opting for repairs rather than full replacements, further diminishing the impact of material wear on device longevity (Makov & Fitzpatrick, 2021).

Functional obsolescence shows a positive and significant coefficient, suggesting a less intuitive relationship where consumers tend to keep smartphones despite performance declines. This aligns with findings by Brey (1997), who observed that users often adapt to reduced functionality, modifying their usage patterns to tolerate minor issues as long as core functionalities remain viable. Consumers may also weigh the inconvenience of replacement against the cost, choosing to delay upgrades until necessary (Cooper, 2004). As Denegri-Knott and Molesworth (2010) suggest, users often wait for the release of new models or discounts, extracting additional value from their current device. This result also resonates with findings by Hahn et al. (2020), who note that consumers may extend device longevity by repurposing or downgrading usage intensity to mitigate performance degradation. Thus, the positive relationship between functional obsolescence and longer lifespans reflects a tendency among consumers to delay replacements until issues become severe or until favourable upgrade opportunities arise.

Psychological obsolescence is negatively and significantly correlated with smartphone lifespan, as expected. Psychological obsolescence refers to the perception of a product as outdated, even when still functional, which drives consumers to replace it (Bulow, 1986; Mugge, Schoormans, & Schifferstein, 2005). In the case of smartphones, the frequent release of newer models, aesthetic shifts, and targeted marketing can render older devices less desirable. This aligns with Jaeger-Erben et al. (2023), who highlight that planned obsolescence is often linked to the perceived status of technology rather than its objective decline. Mugge, Schoormans, and Schifferstein (2005) link aesthetic obsolescence to psychological factors, with consumers replacing products based on appearance alone. For smartphones, this includes slimmer designs and larger screens that appeal to consumer preferences for novelty and social signalling (Swan & Combs, 1976). Finally, economic obsolescence is also important, and the finding is consistent with Bakker et al. (2014), who find that economic factors are particularly relevant for newer models, especially when trade-in or upgrade incentives are available. Along this line of reasoning, Van Nes and Cramer (2005) argue that manufacturer-driven strategies, such as trade-in programs, heighten economic obsolescence by making upgrades appear more cost-effective than repairs, often leveraging both economic and psychological motives to encourage consumers to replace rather than repair. Alzayidi (2024) further confirms this trend, arguing that newer business models actively leverage perceived financial advantages of upgrading, even when devices remain functional.

Beyond these main effects, the analysis also highlights significant interactions between different forms of obsolescence and consumer attitudes. The negative and significant interaction between technological engagement and material obsolescence suggests that individuals with higher technological interest are less affected by material wear, potentially due to their willingness to tolerate minor wear-and-tear as long as the device remains functional. This aligns with previous research indicating that tech-savvy users are more engaged with software capabilities rather than physical attributes when making replacement decisions (Mugge, Schoormans, & Schifferstein, 2005; Prakash et al., 2020; Manier & Mugge, 2022). Similarly, the negative interaction between technological engagement and functional obsolescence implies that users with higher tech interest may adjust to minor performance declines rather than replacing their devices immediately. These consumers might optimize settings, clear storage, or use other workarounds to extend the lifespan of their smartphones despite reduced functionality. In contrast, the positive and highly significant interaction between repair attitude and material obsolescence indicates that consumers who value repairability are much less likely to replace their devices due to material wear, in line with other studies (Makov & Fitzpatrick, 2021). This suggests that promoting repair culture—through policies such as Right to Repair—could mitigate the impact of material degradation on early replacements, encouraging longer smartphone lifespans (Sabbaghi & Behdad, 2018).

Examining the control variables also reveals interesting patterns. As far as the type of activities is concerned, we find a marginally significant coefficient of web_activity on the lifetime of smartphone, suggesting that the intensity and variety of usage reduce the lifetime of smartphones. Interestingly, daily usage appears to be positively associated with the lifespan of smartphones, indicating that heavier daily users may keep their smartphones longer. This might seem counterintuitive, but it aligns with prior discussions on functional obsolescence. First, heavy users are more reliant on their smartphones, often integrating them deeply into their daily routines (e.g., for work, communication, or personal tasks). This reliance can make the transition to a new phone more disruptive, leading to delayed replacement. For example, Van Nes and Cramer (2005) suggest that users with high levels of engagement in using products often exhibit strong product attachment, leading them to invest more in maintaining the device rather than replacing it. High-frequency users may be also more skilled at managing the device's limitations and adapting to minor issues like battery wear or occasional glitches. This is further confirmed by Hahn et al. (2020), who argue that strong user-product attachment contributes to longevity and by Cooper (2005), who discusses how users develop strategies to prolong the life of their devices, such as optimizing phone settings or repairing minor faults instead of opting for replacement. Finally, for heavy users, the perceived cost of replacing a smartphone—both financial and psychological (e.g., data transfer, workflow disruption)—can be higher, leading to a delay in replacement. Baker and Hart (2008) show that the inconvenience of changing devices for technologically embedded users outweighs the functional decline for a considerable time, contributing to longer usage periods.

A last piece of evidence concerns the role of brands. Brand effects are indeed highly significant, with Samsung and other brands showing negative coefficients, indicating that smartphones from these brands tend to have shorter lifespans compared to the reference category - Apple. This suggests that brand perception, durability expectations, and potentially software update policies now play a more influential role in shaping replacement decisions (Jaeger-Erben et al., 2023; Prakash et al., 2020). Consumers may perceive certain brands as offering longer-lasting devices, while others may be associated with faster obsolescence due to performance slow-downs, software update policies, or market positioning.

5.1. Additional evidence: oldest smartphones

To examine potential changes over time in consumer habits and behaviours regarding smartphone replacement, Table 6 presents the estimation results for the lifespan of the oldest smartphone (Smartphone B). As before, Model 1 represents the basic model, Model 2 includes the dummy variables accounting for brands (Apple is the reference case), Model 3 controls for income and Model 4 is the full model including also dummy variables for occupational status.

Table 6
– Variables associated with smartphone's lifetime (recent smartphones).

VARIABLES	Model (1)	Model (2)	Model (3)	Model (4)
age	0.00168	0.00399	0.00968*	0.0154*
	(0.00482)	(0.00480)	(0.00584)	(0.00795)
gender	−0.104	−0.153	−0.204	−0.187
	(0.128)	(0.128)	(0.138)	(0.141)
web_activity	−0.0140	−0.0285	−0.0117	−0.00398
	(0.0514)	(0.0502)	(0.0513)	(0.0513)
filestorage_activity	0.0247	0.0300	0.0507	0.0557
	(0.0704)	(0.0715)	(0.0765)	(0.0778)
dailyusage	−0.00732	−0.00368	−0.00346	−0.00316
	(0.0230)	(0.0227)	(0.0226)	(0.0231)
new	0.698***	0.785***	0.843***	0.864***
	(0.156)	(0.159)	(0.161)	(0.164)
tech_attitude	0.0614	0.0625	0.0724	0.0805*
	(0.0503)	(0.0495)	(0.0492)	(0.0484)
innov_attitude	0.0150	−0.00735	−0.0149	−0.0211
	(0.0558)	(0.0549)	(0.0550)	(0.0549)
repair_attitude	0.0721	0.0956	0.0842	0.0887
	(0.0641)	(0.0643)	(0.0641)	(0.0647)
aesth_attitude	−0.0226	−0.0233	−0.0410	−0.0634
	(0.0562)	(0.0549)	(0.0563)	(0.0581)
techskills	−0.0752	−0.0670	−0.0637	−0.0506
	(0.0554)	(0.0534)	(0.0538)	(0.0519)
materialobs	−0.370*	−0.244	−0.121	−0.146
	(0.201)	(0.206)	(0.191)	(0.223)
functionalobs	0.610***	0.585***	0.631***	0.634***
	(0.212)	(0.209)	(0.217)	(0.217)
psychologicalobs	−0.377***	−0.366***	−0.387***	−0.383***
	(0.138)	(0.139)	(0.136)	(0.142)
economicobs	0.0816	0.0538	−0.0264	−0.0117
	(0.154)	(0.152)	(0.153)	(0.154)
tech_material_interaction	−0.262	−0.295	−0.283	−0.342*
	(0.181)	(0.192)	(0.188)	(0.207)
tech_functional_interaction	−0.0475	−0.0667	−0.0693	−0.111
	(0.162)	(0.156)	(0.145)	(0.141)
repair_material_interaction	1.046*	0.958	0.972*	1.069*
	(0.563)	(0.596)	(0.549)	(0.585)
aesth_psychological_interaction	0.0450	0.0191	0.0718	0.0921
	(0.0747)	(0.0758)	(0.0755)	(0.0757)
Samsung		−0.447***	−0.508***	−0.555***
		(0.142)	(0.142)	(0.148)
Other		−0.409**	−0.440**	−0.475***
		(0.172)	(0.171)	(0.175)
INCOME DUMMY VARIABLES	YES	YES	YES	YES
OCCUPATION DUMMY VARIABLES	YES	YES	YES	YES
Constant	2.430***	2.513***	2.203***	1.832***
	(0.249)	(0.245)	(0.352)	(0.406)
Observations	339	339	339	339
R-squared	0.130	0.158	0.188	0.205

Robust standard errors in parentheses.***p < 0.01, **p < 0.05, *p < 0.1.
Note. All models include dummy variables for income group, occupation, and smartphone brand. For reasons of space and because they are not central to the research questions, only the brand coefficients are reported. The reference category is Apple.

The results highlight a notable shift in consumer behaviour over time regarding smartphone replacement decisions. For older smartphones (Smartphone B), the primary determinant of lifespan was whether the device was purchased new or second-hand, with second-hand devices exhibiting significantly shorter lifespans. However, for newer smartphones (Smartphone A), additional factors have gained importance, suggesting an evolution in consumer awareness and decision-making. While functional obsolescence remains a key driver across both time periods, material obsolescence was more significant for older smartphones, indicating that wear and tear are no longer major triggers for replacement. Instead, consumer engagement with technology, repair attitudes, and psychological factors play a much stronger role for newer smartphones. As mentioned before, technological engagement now significantly moderates the effects of obsolescence, particularly material and functional obsolescence, reflecting that tech-savvy users are more resilient to performance decline and physical deterioration. Additionally, repair-oriented consumers are now much more likely to tolerate material wear, reinforcing the growing relevance of sustainability-driven attitudes (Makov & Fitzpatrick, 2021; Prakash et al., 2020). This shift corresponds to broader market trends, where consumers now demand longer-lasting, repairable products as part of circular economy initiatives (Sabbaghi & Behdad, 2018). Brand effects have also strengthened over time, with Samsung and other brands showing increasingly significant negative impacts on smartphone lifespan, suggesting that brand reputation and consumer trust have become more central to smartphone longevity decisions than in the past.

All in all, these findings suggest that consumers have become more conscious of multiple factors influencing smartphone longevity, moving beyond a purely functional replacement logic. As a result, modern replacement decisions are influenced by a combination of technology engagement, brand perceptions, and repair attitudes—factors that were less relevant in previous smartphone generations. This result can be explained by looking at both technological trends and consumer behaviour. First, technical changes in smartphone design, including better materials, more durable screens, and improved software updates, mean that even second-hand smartphones often perform almost as well as new ones, particularly for recent models. Improvements in product quality can reduce the perceived difference (Bakker et al., 2014; Bayus, 1991) between new and second-hand devices. Second, newer-generation smartphones are often designed with longevity in mind, reducing the performance gap between new and second-hand devices. Besides this, software updates are often supported for several years after release for newer smartphone generations, meaning that second-hand phones can still run the latest operating systems and apps with little functional difference from new smartphones. This reduces the perception that second-hand products are technologically inferior, especially for consumers most interested in the functionality (Brey, 1997). Prakash et al. (2020) further suggest that extended software support has led to a declining rate of perceived obsolescence, particularly among consumers who prioritize efficiency over novelty. Moreover, the rate of innovation in smartphones has slowed, especially regarding improvements in hardware. Many new models have only incremental improvements over previous ones, and this means that even a second-hand phone from the previous generation may offer very similar performance and features to the latest new model, reducing the incentive to choose new over second-hand products. This reflects findings by Valant (2016), who discusses how incremental innovation cycles contribute to longer consumer retention of older models. As discussed in the literature (Cooper, 2005), when the rate of innovation diminishes, the psychological appeal of having the latest model slows down, making second-hand phones more attractive, particularly if they offer near-identical functionality. Alzayidi (2024) confirms this by arguing that the slowing pace of radical innovation has increased consumer resistance to planned obsolescence strategies. Another critical factor explaining our findings is that second-hand products have become more socially acceptable as consumers become more environmentally conscious. At the same time, there is a growing market for certified pre-owned or refurbished phones that come with warranties, making second-hand phones perceived as less risky by consumers. The literature has shown that this evolution in consumers' perceptions - particularly the stigma around second-hand electronics - has been reduced as more consumers seek affordable options and environmental sustainability (Denegri-Knott & Molesworth, 2010). This is reinforced by Jaeger-Erben et al. (2023), who suggest that shifting norms around sustainability have redefined how consumers evaluate second-hand electronics. Finally, second-hand smartphones are becoming a more attractive option for budget-conscious consumers, particularly with newer models being increasingly expensive. The higher upfront cost of new smartphones may push consumers toward second-hand options, where the perceived value is higher for the cost, given that economic efficiency is a crucial determinant of consumers' decision-making (Van Nes & Cramer, 2006; Makov and Fitzpatrick).

6. Conclusions and policy implications

This paper delved into the issue of planned obsolescence, exploring the theoretical reasons that, over the years, encouraged businesses to employ practices that reduce the life cycle of products and the consequences that these dynamics had on the welfare of consumers and on the environment. In particular, the study focused on the smartphone industry, as this sector is widely recognized by both consumers and researchers as being particularly susceptible to planned obsolescence practices (Makov & Fitzpatrick, 2021; Prakash et al., 2020). The decision to focus on smartphones was also supported by their significant environmental impact, given that the majority of their carbon footprint stems from the manufacturing phase. Extending product lifespans and reducing replacement rates, therefore, has major implications for mitigating e-waste and lowering environmental costs (Bocken et al., 2016; Hahn et al., 2020).

The findings reveal that each form of obsolescence plays a significant role, albeit in different ways. Material obsolescence continues to be a strong driver of smartphone replacement, as visible wear and tear often lead to premature replacements, even when the device remains functionally sound. However, this effect appears to be diminishing for newer smartphone generations, where other factors—such as software-related limitations and repairability—play a greater role in replacement decisions (Jaeger-Erben et al., 2023). Functional obsolescence, by contrast, exhibits a positive relationship with smartphone longevity, suggesting that users often adapt to performance issues or delay replacement decisions until absolutely necessary. This finding aligns with prior studies indicating that

consumers, particularly those with high technological engagement, are more resilient to minor performance declines (Brey, 1997; Makov & Fitzpatrick, 2021). Psychological obsolescence, primarily driven by the desire for novelty and social signalling, remains an important determinant, confirming that perceived obsolescence is often just as relevant as technical obsolescence (Mugge, Schoormans, & Schifferstein, 2005; Van Nes & Cramer, 2006). Lastly, economic obsolescence continues to be a key factor, with trade-in programs and rising repair costs encouraging consumers to replace rather than repair their devices (Prakash et al., 2020; Sabbaghi & Behdad, 2018).

The study also identifies a shift in the determinants of smartphone longevity between older and newer models. For older models (Smartphone B), whether the phone was purchased new or second-hand played the most significant role in determining lifespan. Second-hand phones exhibited shorter lifespans, reinforcing earlier findings that used devices are more frequently replaced due to perceived reliability concerns or previous wear (Van Nes & Cramer, 2006). However, for newer models (Smartphone A), this effect diminishes, and additional factors—such as web-based activities, brand perceptions, and repair attitudes—become more relevant. These findings suggest that consumers are becoming more conscious of repairability, sustainability, and technological functionality, rather than simply replacing their devices based on material wear (Alzayidi, 2024; Jaeger-Erben et al., 2023).

6.1. Policy implications

Given these findings, policymakers face an opportunity to address planned obsolescence through regulatory interventions that target the environmental and economic inefficiencies associated with rapid product replacement cycles. Planned obsolescence not only affects consumer welfare but also contributes to environmental degradation and increased greenhouse gas emissions. As such, reducing the frequency of smartphone replacement is not merely a matter of consumer protection, but a strategic lever to support climate policy objectives and the transition to a more circular economy. Regulators are progressively taking a stance to counteract such market failures and guarantee the wellbeing of future generations, while protecting consumers from unfair practices. This aligns with the broader aims of the European Green Deal, which explicitly links product durability and resource efficiency to emissions reduction targets. As this discussion unfolds, it will become clear that the EU is currently leading the way in addressing planned obsolescence, as part of the larger objectives outlined in the Green Deal.

Regarding material obsolescence, policies should promote durability and repairability by encouraging product designs that facilitate easier disassembly and repair. The EU Battery Regulation and Right to Repair initiatives, which compel manufacturers to design batteries that can be easily removed and replaced by consumers, are already steps in this direction (Rovelli, 2023). These measures are not only consumer-oriented but also contribute to reducing the environmental footprint of electronic products by extending their useable life and avoiding premature disposal. However, additional measures like product durability labelling and extended warranties could further empower consumers (European Commission, 2022). These efforts are supported by research highlighting that consumers perceive repairability as a key factor in sustainable choices (Sabbaghi & Behdad, 2018), and that clear repair information could guide more conscious purchasing decisions (Prakash et al., 2020). From an environmental perspective, increased repairability reduces the demand for raw materials and energy-intensive production processes, supporting the decarbonisation goals set at EU level.

Concerning functional obsolescence, regulatory efforts should ensure that consumers receive software updates that maintain device compatibility. The EU Digital Content Directive is an example of how software support can prolong a product's lifecycle (European Commission, 2022). Upgradability requirements for newer models could also mitigate functional obsolescence, allowing users to extend the life of their devices without total replacements. This approach aligns with recent studies showing that software-driven obsolescence remains one of the most subtle yet impactful mechanisms limiting longevity (Makov & Fitzpatrick, 2021). Ensuring longer software support also reduces the carbon footprint associated with premature hardware replacement, as producing and disposing of new smartphones involves substantial emissions across the entire supply chain. Therefore, regulating functional obsolescence is not only a matter of digital consumer rights, but also an environmental imperative.

Regarding psychological obsolescence, which emerges when a product is perceived as ageing or becoming outdated due to subjective factors, policymakers could face challenges in addressing it due to the unquestionable right of companies to innovate and promote their products. Nonetheless, this should not deter governments from addressing this issue through public awareness campaigns and consumer education about the long-term functionality of devices. Furthermore, policymakers could encourage upgradable product designs, which would help reduce the need for replacements driven by aesthetic or novelty factors (European Commission, 2015, 2020). This latter solution is currently being considered by European policymakers as part of a broader initiative aimed to expand the scope of the Ecodesign Directive (European Commission, 2020). Reducing psychological obsolescence is particularly relevant from an environmental standpoint, as frequent and unnecessary replacements accelerate the depletion of resources and increase e-waste volumes. Intervening on social norms, marketing pressures, and consumer expectations may thus complement traditional regulatory tools in reducing the environmental costs of short product life cycles. Alzayidi (2024) argues that firms are beginning to adapt by branding longevity and sustainability as marketable values, which might in turn help reduce the social desirability of frequent replacement.

Finally, to address economic obsolescence, which occurs when the expenses associated with repair and maintenance are higher than the cost of replacement, regulators should promote repairability by mandating access to spare parts and repair manuals for both consumers and independent repairers (Šajn, 2022). Repairability scores, like those introduced in France, and Right-to-Repair legislation can further encourage consumers to repair rather than replace products. Recent moves by companies like Apple, Samsung, and Google to offer self-service repair programs show a potential shift towards embracing repairability and anticipating future regulation (Cosimi, 2022; D'Elia, 2021). Promoting affordable and accessible repair is not only economically beneficial for consumers but also key

to reducing the environmental impact of consumer electronics. Extending product lifetimes through repair directly reduces the need for new device production, which is among the most energy- and resource-intensive phases of the smartphone lifecycle. In this sense, tackling economic obsolescence contributes meaningfully to climate mitigation strategies.

As smartphone lifetimes increase and the market for refurbished devices grows (Malhotra, 2023), the traditional business model of selling new devices may need to evolve. Companies could adopt sustainable business models, such as product-as-a-service approaches, where smartphones are leased rather than sold. These models emphasize long-term service and repairability, shifting the focus from product ownership to product use (Lusch & Vargo, 2014). Modularity and design for upgradability (Bocken et al., 2016; Proske & Jaeger-Erben, 2019) would also help align business models with circular economy principles, allowing users to upgrade components without total replacements.

Moreover, environmental taxation and carbon pricing could indirectly discourage planned obsolescence by incorporating environmental costs into product pricing (European Commission, 2015). This would encourage both manufacturers and consumers to make more sustainable choices, while modular smartphones (e.g., Fairphone) offer valuable case studies of how the industry can transition to more sustainable practices (Wiens, 2022). Modularity could reduce environmental impact across production, use, and disposal phases (Proske & Jaeger-Erben, 2019).

Finally, the progressive shift in consumer preferences towards eco-conscious products poses both challenges and opportunities for manufacturers. Companies can innovate by promoting durability, repairability, and sufficiency in their products, moving away from a consumption-driven paradigm to a more sustainable business model (Bocken et al., 2016). Initiatives such as take-back and refurbish programs (Apple, 2023) show early signs of this transition, though further innovation is necessary to meet the demands of a changing market and regulatory environment.

To conclude, the empirical findings suggest that obsolescence factors—material, functional, psychological, and economic—remain significant drivers of smartphone replacement behaviour. However, as regulatory frameworks evolve and consumer preferences shift towards sustainability, there are opportunities for policymakers to promote durability, repairability, and sufficiency through regulatory and market-based solutions. Additionally, manufacturers can explore new business models that encourage longer product lifetimes and reduce the incentives for planned obsolescence. These models not only support a more circular economy, but also contribute to reducing greenhouse gas emissions by extending product lifetimes and lowering demand for resource-intensive production. Tackling planned obsolescence is thus not only a matter of consumer protection, but also a crucial component of strategies to reduce e-waste, limit environmental impact, and align the electronics sector with climate goals.

6.2. Limitations and future research avenues

This research is subject to several limitations. The first limitation concerns the sample size and structure. Although the survey collected responses from nearly 400 individuals with diverse backgrounds, the study relies on self-reported data, which introduces the potential for response bias. Participants may have provided answers that they perceived as socially desirable or aligned with their values, rather than reflecting their actual behaviours, which could affect the reliability of the results. In relation to this, the sample is skewed toward individuals aged 16–30. While this can limit the external validity, younger individuals tend to be the most intensive and engaged users of smartphones, with high levels of technological proficiency and a greater reliance on mobile devices in their daily lives. Furthermore, younger users are often early adopters of new technologies and are more likely to experience and report on issues related to software updates, hardware degradation, and repairability—factors that are key to understanding the dynamics of obsolescence. As such, their inclusion provides valuable insights into the behavioural and attitudinal drivers of smartphone replacement. While this age concentration may limit generalizability across the entire population, it allows for a focused analysis of the consumer segment most directly affected by, and responsive to, the dynamics of planned obsolescence. In relation to this, we acknowledge also the low explanatory power of our models – which is common in behavioural studies. Given the cross-sectional nature of our data and the relatively small sample size, the objective of our empirical strategy is to identify significant patterns in consumer replacement behaviour, rather than to develop highly predictive models or establish causal relationships.

A further limitation arises from the nature of planned obsolescence itself. As several strategies potentially adopted by manufacturers remain opaque or intentionally undisclosed, it is difficult to access direct and verifiable information about firms' actual practices. This lack of transparency may hinder a comprehensive understanding of the full scope of planned obsolescence in the smartphone industry.

To address these limitations, future research could consider larger and more representative samples, ideally expanding the geographical scope. Comparative studies between European and American consumers, for instance, could explore whether differences in consumption culture and environmental awareness influence smartphone replacement behaviour and attitudes towards obsolescence. Moreover, new research could benefit from qualitative approaches—such as interviews with individuals working within the smartphone industry—to gather insider perspectives on the design, marketing, and support practices that may contribute to planned obsolescence. These insights could validate or challenge public suspicions about how such strategies are deployed in practice. In addition, further studies might employ alternative methodological approaches or analytical frameworks to measure and interpret planned obsolescence. For example, longitudinal research tracking smartphone usage patterns over time, or experimental studies testing consumer responses to changes in repairability, price, or software support, could enrich our understanding of this complex and evolving phenomenon. Finally, future research could further investigate how obsolescence practices shape—and are shaped by—the evolving business models of smartphone manufacturers. This includes exploring the shift towards product-as-a-service schemes, modular design, and circular economy strategies, and how these are influenced by regulatory pressures and changing consumer expectations.

CRediT authorship contribution statement

Nicoletta Corrocher: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Sara Paganuzzi:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

APPENDIX

Table A1
– Principal component analysis on all the constructs

Factor 1 – INNOVATION		Factor 2 - AESTHETICS	
techfundamental	−0.0267	innovationreplacement	0.0205
repairifdamaged	0.0801	techtool	0.0298
techtool	0.1626	repairindependentshops	0.0308
repairnoifnewphonecheaper	−0.1953	repairifdamaged	0.0577
repairindependentshops	−0.1981	techintegration	0.0704
innovationcamera	0.3256	innovationnewversion	0.1024
techthreat	0.3312	innovationknowledge	0.1533
aestheticcracks	0.3343	techfundamental	0.1759
techintegration	0.4068	aestheticpeers	0.1978
aestheticdesign	0.4434	techthreat	−0.2169
aestheticstyle	0.4989	aestheticcracks	0.2791
innovationfeatures	**0.5680**	aestheticold	0.3897
aestheticold	0.6218	**innovationfeatures**	**0.5415**
innovationreplacement	**0.7099**	**innovationcamera**	**0.5931**
innovationknowledge	**0.7807**	**aestheticstyle**	**0.5993**
aestheticpeers	**0.7949**	**repairnoifnewphonecheaper**	**0.6264**
innovationnewversion	**0.8059**	**aestheticdesign**	**0.6600**
Factor 3 - TECHNOLOGY		**Factor 4 - REPAIR**	
aestheticold	0.0142	techfundamental	−0.0135
innovationreplacement	0.0192	innovationfeatures	0.0138
repairindependentshops	−0.0268	aestheticstyle	−0.0150
aestheticstyle	−0.0381	innovationnewversion	−0.0229
aestheticdesign	−0.0397	aestheticpeers	−0.0237
innovationcamera	0.0489	repairnoifnewphonecheaper	0.0251
aestheticpeers	0.1148	innovationcamera	0.0501
innovationnewversion	0.1231	innovationknowledge	−0.0583
aestheticcracks	0.1231	techtool	0.0663
innovationknowledge	0.1256	aestheticdesign	0.0975
innovationfeatures	0.1637	techthreat	0.1329
repairifdamaged	0.1705	aestheticcracks	0.1331
repairnoifnewphonecheaper	0.3791	innovationreplacement	−0.1748
techintegration	**0.5436**	aestheticold	0.2168
techthreat	**0.5550**	techintegration	0.2275
techtool	**0.7203**	**repairifdamaged**	**0.7532**
techfundamental	**0.7805**	**repairindependentshops**	**0.7653**

Cronbach Alpha: 0.8465.

Data availability

Data will be made available on request.

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